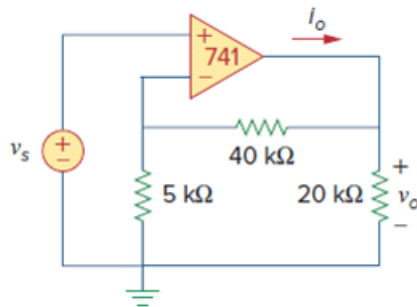


Problem

If the same 741 op amp in Example 5.1 is used in the circuit of Fig. 5.7, calculate the closed-loop gain v_o/v_s . Find i_o when $v_s = 1$ V.

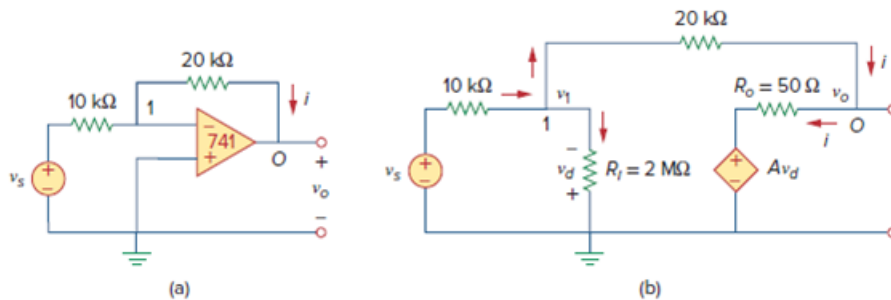
Figure. 5.7:



Example 5.1:

A 741 op amp has an open-loop voltage gain of 2×10^5 , input resistance of $2 \text{ M}\Omega$, and output resistance of 50Ω . The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s . Determine current i when $v_s = 2$ V.

Figure. 5.6: (a) original circuit, (b) the equivalent circuit.



Solution:

Using the op amp model in Fig. 5.4, we obtain the equivalent circuit of Fig. 5.6(a) as shown in Fig. 5.6(b). We now solve the circuit in Fig. 5.6(b) by using nodal analysis. At node 1, KCL gives

$$\frac{v_s - v_1}{10 \times 10^3} = \frac{v_1}{2000 \times 10^3} + \frac{v_1 - v_o}{20 \times 10^3}$$

Multiplying through by 2000×10^3 , we obtain

$$200v_s = 301v_1 - 100v_o$$

Or

$$2v_s \approx 3v_1 - v_o \Rightarrow v_1 = \frac{2v_s + v_o}{3} \quad (5.1.1)$$

At node O,

$$\frac{v_1 - v_o}{20 \times 10^3} = \frac{v_o - Av_d}{50}$$

But $v_d = -v_1$ and $A = 200,000$. Then

$$v_1 - v_o = 400(v_o + 200,000v_1) \quad (5.1.2)$$

Substituting v_1 from Eq. (5.1.1) into Eq. (5.1.2) gives

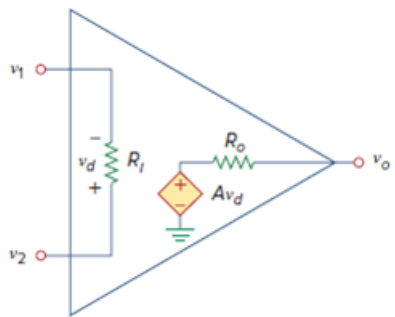
$$0 \approx 26,667,067v_o + 53,333,333v_s \Rightarrow \frac{v_o}{v_s} = -1.9999699$$

This is closed-loop gain, because the $20\text{-k}\Omega$ feedback resistor closes the loop between the output and input terminals. When $v_s = 2$ V, $v_o = -3.9999398$ V. From Eq. (5.1.1), we obtain $v_1 = 20.066667 \mu\text{V}$. Thus,

$$i = \frac{v_1 - v_o}{20 \times 10^3} = 0.19999 \text{ mA}$$

It is evident that working with a nonideal op amp is tedious, as we are dealing with very large numbers.

Figure. 5.4: The equivalent circuit of the nonideal op amp.

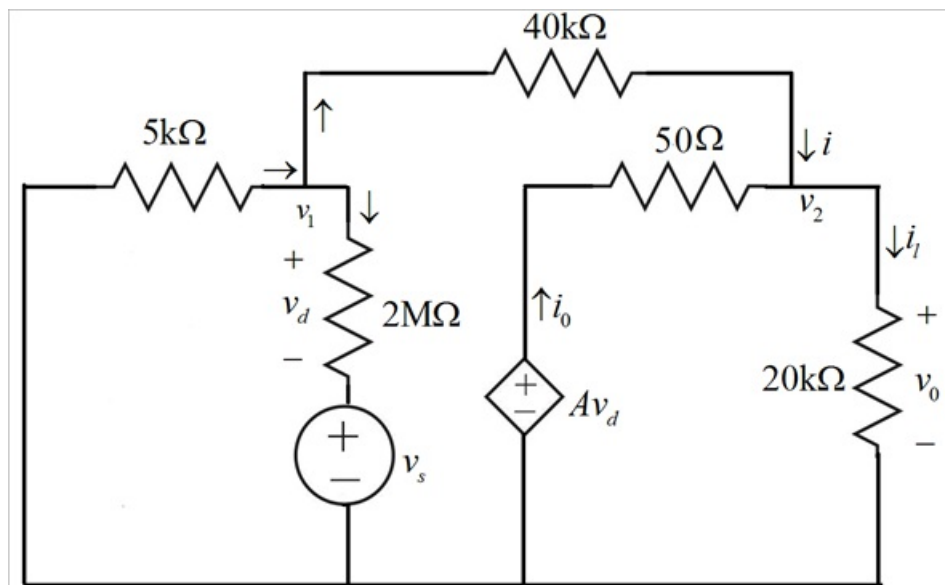


Step-by-step solution

Step 1 of 5

Refer to Figure 5.7 in the textbook.

Draw the equivalent circuit of an op-amp.



Step 2 of 5

Apply Kirchhoff's current law at node v_1 .

$$\frac{v_1 - v_s}{2 \times 10^6} + \frac{v_1}{5 \times 10^3} + \frac{v_1 - v_2}{40 \times 10^3} = 0$$

$$v_1 \left(\frac{1}{2 \times 10^6} + \frac{1}{5 \times 10^3} + \frac{1}{40 \times 10^3} \right) = \frac{v_s}{2 \times 10^6} + \frac{v_2}{40 \times 10^3}$$

$$2.255 \times 10^{-4} v_1 = \frac{v_s}{2 \times 10^6} + \frac{v_2}{40 \times 10^3}$$

$$v_1 = \frac{v_s}{451} + \frac{v_2}{9.02} \dots\dots (1)$$

Step 3 of 5

Apply Kirchhoff's voltage law to the first loop.

$$-v_1 + v_d + v_s = 0$$

$$v_d = -v_s + v_1$$

Apply Kirchhoff's current law at node v_2 .

$$\frac{v_2 - Av_d}{50} + \frac{v_2 - v_1}{40 \times 10^3} + \frac{v_2}{20 \times 10^3} = 0$$

$$v_2 \left(\frac{1}{50} + \frac{1}{40 \times 10^3} + \frac{1}{20 \times 10^3} \right) = \frac{Av_d}{50} + \frac{v_1}{40 \times 10^3}$$

Substitute 200000 for gain A .

$$0.020075v_2 = \frac{200000(-v_s + v_1)}{50} + \frac{v_1}{40 \times 10^3}$$

$$0.020075v_2 = -4000v_s + 4000v_1 + 2.5 \times 10^{-5}v_1 \dots\dots (2)$$

Step 4 of 5

Calculate the closed loop gain.

Substitute the value for v_1 from equation (1).

$$0.020075v_2 = -4000v_s + 4000.000025 \left(\frac{v_s}{451} + \frac{v_2}{9.02} \right)$$

$$0.020075v_2 = -4000v_s + 8.869179656v_s + 443.4589828v_2$$

$$443.4389078v_2 = 39991.13082v_s$$

$$\frac{v_2}{v_s} = 9.00041$$

From the circuit,

$$v_2 = v_0$$

$$\frac{v_0}{v_s} = 9.00041$$

Therefore, the closed loop gain is 9.00041.

Step 5 of 5

Determine the value of i_0 .

From the circuit,

$$i_0 = i_1 + i$$

$$i_0 = \frac{v_0}{20 \times 10^3} + \frac{v_0 - v_1}{40 \times 10^3}$$

Calculate the value of v_1 from equation (2).

Substitute 1 V for v_s , 9.00041 V for v_2 .

Therefore, replace 1.000045455 for v_1 .

$$\begin{aligned} i_0 &= \frac{9.00041}{20 \times 10^3} + \frac{9.00041 - 1.000045455}{40 \times 10^3} \\ &= (4.500205 \times 10^{-4}) + (2.000091136 \times 10^{-4}) \\ &= 6.5 \times 10^{-4} \end{aligned}$$

Therefore, the value of i_0 is 0.65mA.